Introduction

When Apress approached me, several years ago, with the idea of writing my own book, I really did not know what I was letting myself in for. At that point I had been the technical reviewer for two editions of Matt Zandstra's *PHP 8 Objects, Patterns, and Practice* but had little experience of writing otherwise. For unfathomable reasons, Apress floated the idea of becoming an author and pitched a title to me, which I fleshed out to a full proposal. I settled in and began to write. That book failed utterly to spring forth from my mind, and instead something else began to emerge: the book you are reading now.

I realized quite early on in the writing process that what I wanted to do more than anything was to teach the *why* as well as the *how*. My reasoning was that the best thing I can do for you, the reader, is to show what is happening behind the curtain. This book therefore contains many gory details on how the machines work and how they talk to each other, because this is what informs the design choices of everything else that is layered on top – including PHP.

This book makes no assumptions about your level of understanding of computing in general, other than that you know how to operate one to a certain degree. We start with the basics of setting yourself up with an environment you can use to follow the myriad of examples contained within, as well as the fundamentals of programming and PHP syntax – the more experienced reader can, of course, skim over these parts. Once these foundational topics are covered, the book goes on to dive quite deeply into how things *really* work behind the scenes, especially text data (which is arguably the chief concern of web development).

Therefore, I think that even seasoned developers will find a lot of value within these pages, especially if (like me) you are self-taught – as so many of us jobbing web devs are. Over the years (decades!) I found that I was often descending down rabbit holes of computer science and web standards to fill in the gaps in my knowledge, and the fruit of those late-night reading and hacking sessions is presented here. May it serve you well.

Things change all the time in web development: frameworks, tools, even languages… they come and go like the seasons. Other things, however, stand the test of time. The protocols underpinning the web, and the machines doing the work, haven't changed all that much in the decades since a smart chap at CERN thought it would be great if